

TEERTHANKER MAHAVEER UNIVERSITY – MORADABAD

B.Tech VI (Sixth) Semester Examination 2023-24

Course Code: EAI602

Paper ID: 09624626

Genetic Algorithms

Time: 3 Hours

Max. Marks: 60

Note: Attempt **all** questions.

Q1.	Answer the following (limit your answer to 50 words):	Marks
a)	Define the term 'genes'.	2
	Or	
	Explain about biological neuron.	2
b)	What is search space in Genetic Algorithm?	2
	Or	
	Explain about the applications of random search.	2
c)	Differentiate between permutation encoding and value based encoding.	2
	Or	
	Why use stochastic universal sampling?	2
d)	Define the various application of genetic algorithm.	2
	Or	
	What is Genetic-Neurohybrid system.	2
e)	What is flipping mutation?	2
	Or	
	Differentiate between flipping mutation and interchanging mutation.	2
2.	How Genetic Algorithm is different from traditional search algorithm? Write limitation of GA.	10
	Or	
	Examine the fundamental model of artificial neural network.	10
3.	You're tuning a machine learning model. How can stochastic hill climbing find good settings without trying every possibility?	10
	Or	
	Using genetic algorithms maximize $f(X) = x^2$ over $\{0, 1, 2, \dots, 31\}$ with initial X values of $\{13, 24, 8, 16\}$. Show one crossover and mutation operation.	10
4.	Identify the basic operators of genetic algorithm? Explain the operational procedure of GA with example.	10
	Or	
	Apply Binary encoding, octal encoding, hexadecimal encoding in GA with suitable examples.	10
5.	Define the terms fitness function, crossover and mutation as used in genetic algorithms. Explain how genetic algorithms work, in English or in pseudocode.	10
	Or	
	Explain different crossover operations performed in genetic algorithm.	10

6.	What is convergence? Explain the stopping criteria in genetic algorithm.	10
	Or	
	Explain the step by step procedure for solving economic dispatch problems using genetic algorithm.	10

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Q1.	Answer the following (limit your answer to 50 words):	Marks
a)	What is genetic programming and how does it differ from a traditional genetic algorithm (GA)	2
	Or	
	Explain about the Mc Culloch-Pitts neuron model.	2
b)	Compare artificial neural network and biological neural network.	2
	Or	
	Define the role of population in evolutionary algorithm?	2
c)	Define the various applications of biological neural network.	2
	Or	
	Compare single point crossover and two point crossover operator.	2
d)	What do you understand by genetic algorithm in machine learning?	2
	Or	
	Why use stochastic universal sampling?	2
e)	What do you understand by biological neuron?	2
	Or	
	What are the types of neuron activation functions?	2
2.	Explain about the basic models of artificial neural networks.	10
	Or	
	Analyze Genetic Algorithm is different from traditional search algorithm? Write the benefits for genetic algorithm.	10
3.	Make use of different reproduction operators used in genetic algorithm with suitable example.	10
	Or	
	Construct some attractive features of the biological neural network that make it superior to the most sophisticated Artificial Intelligence computer system for pattern recognition tasks.	10
4.	Construct at least one practical example on any five below mentioned topics: a) Binary Encoding b) OCTAL Encoding c) Hexadecimal encoding d) Value encoding e) tree encoding f) permutation encoding	10
	Or	
	Build different mutation operations performed in genetic algorithm.	10
5.	Define the terms fitness function, crossover and mutation as used in genetic algorithms.	10

	Or	
	Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a + b) - (c + d) + (e + f) - (g + h)$ and let the initial population consist of four individuals with the following chromosomes: $\begin{aligned} x_1 &= 6\ 5\ 4\ 1\ 3\ 5\ 3\ 2 \\ x_2 &= 8\ 7\ 1\ 2\ 6\ 6\ 0\ 1 \\ x_3 &= 2\ 3\ 9\ 2\ 1\ 2\ 8\ 5 \\ x_4 &= 4\ 1\ 8\ 5\ 2\ 0\ 9\ 4 \end{aligned}$ Show the fitness of each individual, showing all your workings, and arrange them in order with the fittest first and the least fit last.	10
6.	Explain the step by step procedure for solving economic dispatch problems using genetic algorithm.	10
	Or	
	What is convergence? Explain the termination criteria in genetic algorithm	10

Course Code: EAI602	Professional Core Course-16 Specialization- Artificial Intelligence, Machine Learning & Deep Learning B.Tech (CSE) Semester VI GENETIC ALGORITHMS	L-3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO1.	Understanding the essential features of genetic algorithm (GA) and to evaluate the population, fitness and search space in it.	
CO2.	Understanding the concepts of encoding, decoding in genetics and implement the various operators and features of GA.	
CO3.	Applying the optimization and searching techniques in search space.	
CO4.	Applying GA for building solutions to various problems and to study and evaluate the stopping criteria for the algorithm.	
CO5.	Analyzing and applying different crossover and mutation operators for effectively solving the desired real-world problems.	
Course Content:		
Unit-1:	Introduction of Genetic Algorithm: Objective, learning objectives, Biological correlation with ANN: cell, chromosomes, genes, Population, fitness, reproduction, creation of Offspring, search space, working principle-Simple GA.	8 Hours
Unit-2:	Optimization & Search Technique: Gradient Based local optimization Method, Random Search, Stochastic Hill Climbing, Simulated Annealing, Symbolic Artificial Intelligence. Genetic Algorithm & Search Space: Search Space, world of Genetic Algorithm, Evolution & Optimization, Evolution & GA, Genetic Algorithm v/s Traditional Algorithm.	8 Hours
Unit-3:	Genetic Modelling: Encoding of GA-Binary Encoding, OCTAL Encoding, Hexadecimal encoding, value encoding, tree encoding and permutation encoding, Fitness function. Operators in GA: (Selection (Reproduction), Crossover, Mutation. Selection (Reproduction): Roulette wheel Selection, Random Selection, Rank Selection, Boltzmann Selection, Elitism, Stochastic universal Sampling.	8 Hours
Unit-4:	Crossover Operators: Single-Point Crossover, Two-Point Crossover, N-point Crossover, Uniform Crossover, three parent Crossover, Crossover with reduced surrogate, Shuffle Crossover, Ordered Crossover. Mutation Operator: introduction of Mutation, Flipping Mutation, Interchanging Mutation, Reversing Mutation.	8 Hours
Unit-5:	Stopping Criteria of GA, Various constraints of GA, Problem Solving Using Genetic Algorithm, Various applications of GA. Genetic- Neuro Hybrid System.	8 Hours
Text Books:	1. S. N. Sivanandam& S. N. Deepa, Principles of S.oft Computing, 2nd Edition, Willy Publication.	
Reference Books:		